MPEC Storage nachvollziehen

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A. Sets and Indices

~~B~~	~~Set of buses, indexed by b.~~ → no bus
I	Set of thermal generators, indexed by i.
~~L~~	~~Set of transmission lines, indexed by l.~~ → no lines!
H	Set of storage devices, indexed by h.
~~W~~	~~Set of wind generators, indexed by w.~~ treated as gen
~~Ω_1~~	~~Set of wind scenarios at D_1, indexed by ω_1.~~
~~Ω~~	~~Set of wind scenarios, indexed by ω. Note that ω_0 stands for D_0 deterministic forecast profiles.~~
T^{D_0}	Set of time intervals of the first market window indexed by $t_0 = 1 \dots N_{T_0}$.
T^{D_1}	Set of time intervals of the second market window indexed by $t_1 = (N_{T_0}+1) \dots N_{T_1}$.
T	Set of all time intervals, indexed by t. Note that $T = T^{D_0} \cup T^{D_1}$.
~~$f(l)/t(l)$~~	~~Indices of the from/to buses of line l.~~

B. Variables

$\mathrm{chs}_{h,t}/\mathrm{dis}_{h,t}$	Quantity of charging bid/discharging offer of storage h at time t, MW.
$\rho^{\mathrm{chs}}_{h,t}/\rho^{\mathrm{dis}}_{h,t}$	Price of charging bid/discharging offer of storage h at time t, \$/MWh.
$x^{\mathrm{dis}}_{h,t}$	Binary status variable of storage h at time t. Equals to 1 if discharging and 0 otherwise.
$\Pi^{D_0}/\Pi^{D_1}_{\omega_1}$	Profit at D_0 / D_1 stage.
~~Γ^{D_1}~~	~~Risk-constrained look-ahead profits adjustments.~~
~~CVaR^{D_1}~~	~~Conditional value-at-risk of D_1 profits.~~
~~VaR^{D_1}~~	~~Value-at-risk of D_1 profits.~~
~~$\mathrm{loss}^{D_1}_{\omega_1}$~~	~~Auxiliary variables to compute D_1 CVaR.~~
$\mathrm{SoC}_{h,t,\omega}$	State of charge of storage h at time t and scenario ω, MWh.
$d_{b,t,\omega}$	Cleared demand at bus b at time t and scenario ω, MW.
$p^G_{i,t,\omega}$	Cleared power output of conventional generator i at time t and scenario ω, MW.
~~$p^W_{w,t,\omega}$~~	~~Cleared power output of wind generator w at time t and scenario ω, MW.~~
~~$pf_{l,t,\omega}$~~	~~Power flow on line l at time t and scenario ω, MW.~~
$q^{\mathrm{dis/chs}}_{h,t,\omega}$	Cleared discharging/charging rate for storage h at time t and scenario ω, MW.
~~$\theta_{b,t,\omega}$~~	~~Voltage phase angle at bus b at time t and scenario ω, rad.~~

C. Dual Variables

$\lambda_{b,t,\omega}$	Nodal power balance constraint at time t and scenario ω.
$\underline{\mu}^G_{i,t,\omega}/\bar{\mu}^G_{i,t,\omega}$	Min/max power output constraints of generator i at time t and scenario ω.
~~$\underline{\mu}^W_{w,t,\omega}/\bar{\mu}^W_{w,t,\omega}$~~	~~Min/max power output constraints for wind generator w at time t and scenario ω.~~
$\underline{\mu}^D_{b,t,\omega}/\bar{\mu}^D_{b,t,\omega}$	Min/max bounds on demand at bus b time t at scenario ω.
$\underline{\alpha}^{\mathrm{dis}}_{h,t,\omega}/\bar{\alpha}^{\mathrm{dis}}_{h,t,\omega}$	Min/max discharging rate constraints of storage h at time t and scenario ω.
$\underline{\alpha}^{\mathrm{chs}}_{h,t,\omega}/\bar{\alpha}^{\mathrm{chs}}_{h,t,\omega}$	Min/max charging rate constraints of storage h at time t and scenario ω.
~~$\nu_{l,t,\omega}$~~	~~Power flow constraints on line l at time t and scenario ω.~~

D. Parameters

~~γ^{D_1}~~	~~Discount factor for the profit term of day D_1.~~
~~γ^{CVaR}~~	~~Weight factor for the risk position of day D_1.~~
C^{ES}_h	Variable operating cost for storage h, \$/MW.
C^G_i/C^W_w	Offer price for generator i / wind w, \$/MWh.
C^D_b	Bid price for demand b, \$/MWh.
$\overline{dis}_h/\overline{chs}_h$	Discharging/charging rate limit for storage h, MW.
$\eta^{\mathrm{dis}}_h/\eta^{\mathrm{chs}}_h$	Discharging/charging efficiency for storage h.
$\underline{D}_{b,t}/\overline{D}_{b,t}$	Min/max bounds on demand at bus b at time t, MW.
~~$\bar{F}_l$~~	~~Power flow limit on line l, MW.~~
$\underline{P}_i/\overline{P}_i$	Min/max power output limits for generator i, MW.
$P^G_{i,0}$	Initial power output of generator i, MW.
RU_i/RD_i	Ramp up/down limits for generator i, MW.
$\underline{\mathrm{SoC}}_h/\overline{\mathrm{SoC}}_h$	Min/max state-of-charge for storage h, MWh.
$\mathrm{SoC}^{\mathrm{init}}_h$	Initial state-of-charge level for storage h, MWh.
~~X_l~~	~~Reactance of line l.~~
~~π_{ω_1}~~	~~Probability for wind scenario ω_1 on day D_1.~~
~~β_0~~	~~Probability threshold for risk calculation.~~
~~$A_{l,b}$~~	~~Incidence matrix of transmission lines. $b=1$ indicates bus b is the from/to bus for line l, and 0 otherwise.~~
~~$WF_{w,t,\omega}$~~	~~Forecast wind w at time t and scenario ω, MW.~~

Upper level Problem: Strategic Storage Decision

$$\max \quad Obj = \Pi^{D_0} + \cancel{\gamma^{D_1}\Gamma^{D_1}} \qquad (1)$$

profit; risk-shit; primary objective

subject to:

$$\cancel{\Gamma^{D_1} = \gamma^{CVaR} CVaR^{D_1} + (1-\gamma^{CVaR}) \sum_{\omega_1} (\pi_{\omega_1} \Pi^{D_1}_{\omega_1})} \qquad (2)$$

$$\cancel{loss^{D_1}_{\omega_1} \geq VaR^{D_1} - \Pi^{D_1}_{\omega_1}, \forall \omega_{1,}} \qquad (3)$$

$$\cancel{loss^{D_1}_{\omega_1} \geq 0, \forall \omega_{1,}} \qquad (4)$$

$$\cancel{CVaR^{D_1} = VaR^{D_1} - \frac{1}{1-\beta_0} \sum_{\omega_1} (\pi_{\omega_1} loss^{D_1}_{\omega_1})} \qquad (5)$$

$$\Pi^{D_0} = \sum_{h,t_0} [\lambda_{b(h),t_0} (q^{\mathrm{dis}}_{h,t_0} - q^{\mathrm{chs}}_{h,t_0}) - C^{ES}_h (q^{\mathrm{dis}}_{h,t_0} + q^{\mathrm{chs}}_{h,t_0})] \qquad (6)$$

market price; storage mc 4 Shit; profit definition

$$\Pi^{D_1}_{\omega_1} = \sum_{h,t_1} [\lambda_{b(h),t_1,\omega_1} (q^{\mathrm{dis}}_{h,t_1,\omega_1} - q^{\mathrm{chs}}_{h,t_1,\omega_1})]$$

second market I do not care

I am abstracting from this so full information across all modelled t assumed

$$- C_h^{ES}(q^{\text{dis}}_{h,t_0} + q^{\text{chs}}_{h,t_0})] \quad (6)$$

4 ESM

second market / do not care about

$$\cancel{\Pi^{D_1}_{\omega_1} = \sum_{h,t_1} [\lambda_{b(h),t_1,\omega_1}(q^{\text{dis}}_{h,t_1,\omega_1} - q^{\text{chs}}_{h,t_1,\omega_1})}$$

$$\cancel{- C_h^{ES}(q^{\text{dis}}_{h,t_1,\omega_1} + q^{\text{chs}}_{h,t_1,\omega_1})] \quad (7)}$$

I am abstracting from this so full information across all modelled t assumed

$$0 \le chs_{h,t} \le \overline{chs}_h(1 - x^{\text{dis}}_{h,t}), \forall h, \forall t \quad (8)$$

$$0 \le dis_{h,t} \le \overline{dis}_h x^{\text{dis}}_{h,t}, \forall h, \forall t \quad (9)$$

ensures either charge or discharge within bounds of charge & discharge power

$$x^{\text{dis}}_{h,t} \in \{0,1\}, \forall h, \forall t \quad (10)$$

binary for discharging / charging

$$\rho^{\text{dis}}_{h,t}, \rho^{\text{chs}}_{h,t} \ge 0, \forall h, \forall t \quad (11)$$

prices

do not allow negative storage bids!

$$SoC_{h,t_0} = SoC^{init}_h + q^{\text{chs}}_{h,t_0}\eta^{\text{chs}}_h - q^{\text{dis}}_{h,t_0}/\eta^{\text{dis}}_h,$$

$$\forall t_0 = 1, \forall h \quad (12)$$

$$SoC_{h,t_0} = SoC_{h,t_0-1} + q^{\text{chs}}_{h,t_0}\eta^{\text{chs}}_h - q^{\text{dis}}_{h,t_0}/\eta^{\text{dis}}_h,$$

$$\forall 1 < t_0 \le N_{t_0}, \forall h \quad (13)$$

SoC updates

$$\underline{SoC}_h \le SoC_{h,t_0} \le \overline{SoC}_h, \forall h, \forall t_0 \quad (14)$$

$$\cancel{SoC_{h,t_1,\omega_1} = SoC_{h,N_{t_0}} + q^{\text{chs}}_{h,t_1,\omega_1}\eta^{\text{chs}}_h - q^{\text{dis}}_{h,t_1,\omega_1}/\eta^{\text{dis}}_h, \forall t_1 = N_{t_0}+1, \forall h, \forall \omega_1 \quad (15)}$$

$$\cancel{SoC_{h,t_1,\omega_1} = SoC_{h,t_1-1,\omega_1} + q^{\text{chs}}_{h,t_1,\omega_1}\eta^{\text{chs}}_h - q^{\text{dis}}_{h,t_1,\omega_1}/\eta^{\text{dis}}_h, \forall N_{t_0}+1 < t_1 \le N_{t_1}, \forall h, \forall \omega_1 \quad (16)}$$

$$\cancel{\underline{SoC}_h \le SoC_{h,t_1,\omega_1} \le \overline{SoC}_h, \forall h, \forall t_1, \forall \omega_1 \quad (17)}$$

$$\cancel{SoC_{h,N_{t_1},\omega_1} \ge SoC^{init}_h, \forall h, \forall \omega_1 \quad (18)}$$

Lower Level Problem: Market Clearing

subject to:

$$\lambda_{b,t,\omega}, q^{\text{dis}}_{h,t,\omega}, q^{\text{chs}}_{h,t,\omega} \in \arg\max \Big\{ SW_\omega = \sum_{b,t} C^D_b\, d_{b,t,\omega}$$

demand costs

$$+ \sum_{h,t} \rho^{\text{chs}}_{h,t}\, q^{\text{chs}}_{h,t,\omega} - \sum_{i,t} C^G_i\, p^G_{i,t,\omega} - \sum_{h,t} \rho^{\text{dis}}_{h,t}\, q^{\text{dis}}_{h,t,\omega} \quad (19)$$

charging costs; generators price; discharging costs

Wohlfahrtsoptimum

$$d_{b,t,\omega} + \sum_{f(l)=b} pf_{l,t,\omega} - \sum_{t(l)=b} pf_{l,t,\omega} = \sum_{b(i)=b} p^G_{i,t,\omega}$$

$$+ \sum_{b(w)=b} p^W_{w,t,\omega} + \sum_{b(h)=b} (q^{\text{dis}}_{h,t,\omega} - q^{\text{chs}}_{h,t,\omega}) : (\lambda_{b,t,\omega}), \forall b, \forall t \quad (20)$$

Energy Balance

$$0 \le p^G_{i,t,\omega} \le \overline{P}_i : (\underline{\mu}^G_{i,t,\omega}, \bar{\mu}^G_{i,t,\omega}), \forall i, \forall t \quad (21)$$

$$0 \le p^W_{w,t,\omega} \le WF_{w,t,\omega} : (\underline{\mu}^W_{w,t,\omega}, \bar{\mu}^W_{w,t,\omega}), \forall w, \forall t \quad (22)$$

$$\underline{D}_{b,t} \le d_{b,t,\omega} \le \overline{D}_{b,t} : (\underline{\mu}^D_{b,t,\omega}, \bar{\mu}^D_{b,t,\omega}), \forall b, \forall t \quad (23)$$

$$0 \le q^{\text{dis}}_{h,t,\omega} \le dis_{h,t} : (\underline{\alpha}^{\text{dis}}_{h,t,\omega}, \bar{\alpha}^{\text{dis}}_{h,t,\omega}), \forall h, \forall t \quad (24)$$

$$0 \le q^{\text{chs}}_{h,t,\omega} \le chs_{h,t} : (\underline{\alpha}^{\text{chs}}_{h,t,\omega}, \bar{\alpha}^{\text{chs}}_{h,t,\omega}), \forall h, \forall t \quad (25)$$

min & max power limits of bids

$$p^G_{i,t,\omega} - p^G_{i,t-1,\omega} \le RU_i : (\beta^u_{i,t,\omega}), \forall t > 1, \forall i \quad (26)$$

$$p^G_{i,t-1,\omega} - p^G_{i,t,\omega} \le RD_i : (\beta^d_{i,t,\omega}), \forall t > 1, \forall i \quad (27)$$

$$p^G_{i,t,\omega} - p^G_{i,0} \le RU_i : (\beta^u_{i,t,\omega}), t = 1, \forall i \quad (28)$$

$$p^G_{i,0} - p^G_{i,t,\omega} \le RD_i : (\beta^d_{i,t,\omega}), t = 1, \forall i \quad (29)$$

ramping constraints of generators

$$pf_{l,t,\omega} = \frac{1}{X_l}(\theta_{f(l),t,\omega} - \theta_{t(l),t,\omega}) : (\nu_{l,t,\omega}), \forall l, \forall t \quad (30)$$

$$\cancel{-\overline{F}_l \le pf_{l,t,\omega} \le \overline{F}_l : (\underline{\varphi}_{l,t,\omega}, \bar{\varphi}_{l,t,\omega}), \forall l, \forall t \quad (31)}$$

$$\cancel{-\bar{\theta} \le \theta_{b,t,\omega} \le \bar{\theta} : (\underline{\delta}_{b,t,\omega}, \bar{\delta}_{b,t,\omega}), \forall b \ne \text{ref}, \forall t \quad (32)}$$

$$\cancel{\theta_{b,t,\omega} = 0 : (\delta^{\text{ref}}_t), b = \text{ref}, \forall t \Big\} \forall \omega \quad (33)}$$

power flow makes no sense

KKT $\Longrightarrow$

$$C^G_i - \lambda_{b(i),t,\omega} - \underline{\mu}^G_{i,t,\omega} + \bar{\mu}^G_{i,t,\omega} + \beta^u_{i,t,\omega} - \beta^d_{i,t,\omega}$$

$$+ \beta^d_{i,t+1,\omega|t<N_{T_1}} - \beta^u_{i,t+1,\omega|t<N_{T_1}} = 0, \forall i, \forall t \quad (34)$$

$$-\lambda_{b(w),t,\omega} - \underline{\mu}^W_{w,t,\omega} + \bar{\mu}^W_{w,t,\omega} = 0, \forall w, \forall t \quad (35)$$

$$-C^D_b + \lambda_{b,t,\omega} - \underline{\mu}^D_{b,t,\omega} + \bar{\mu}^D_{b,t,\omega} = 0, \forall b, \forall t \quad (36)$$

$$\rho^{\text{dis}}_{h,t} - \lambda_{b(h),t,\omega} - \underline{\alpha}^{\text{dis}}_{h,t,\omega} + \bar{\alpha}^{\text{dis}}_{h,t,\omega} = 0, \forall h, \forall t \quad (37)$$

$$-\rho^{\text{chs}}_{h,t} + \lambda_{b(h),t,\omega} - \underline{\alpha}^{\text{chs}}_{h,t,\omega} + \bar{\alpha}^{\text{chs}}_{h,t,\omega} = 0, \forall h, \forall t \quad (38)$$

$$\cancel{\sum_b A_{l,b}\lambda_{b,t,\omega} - \underline{\varphi}_{l,t,\omega} + \bar{\varphi}_{l,t,\omega} - \nu_{l,t,\omega} = 0, \forall l, \forall t \quad (39)}$$

$$\cancel{\sum_l \frac{A_{l,b}\nu_{l,t,\omega}}{X_l} - \underline{\delta}_{b,t,\omega} + \bar{\delta}_{b,t,\omega} = 0, \forall b \ne \text{ref}, \forall t \quad (40)}$$

$$\cancel{\sum_l \frac{A_{l,b}\nu_{l,t,\omega}}{X_l} - \delta^{\text{ref}}_{t,\omega} = 0, b = \text{ref}, \forall t \quad (41)}$$

dual feasibility

power flow

$$\underline{\mu}^G_{i,t,\omega} p^G_{i,t,\omega} = 0, \forall i, \forall t \quad (42)$$

$$\bar{\mu}^G_{i,t,\omega}(\overline{P}_i - p^G_{i,t,\omega}) = 0, \forall i, \forall t \quad (43)$$

$$\underline{\mu}^W_{w,t,\omega} p^W_{w,t,\omega} = 0, \forall w, \forall t \quad (44)$$

$$\bar{\mu}^W_{w,t,\omega}(WF_{w,t,\omega} - p^W_{w,t,\omega}) = 0, \forall w, \forall t \quad (45)$$

$$\underline{\mu}^D_{b,t,\omega}(d_{b,t,\omega} - \underline{D}_{b,t}) = 0, \forall b, \forall t \quad (46)$$

$$\bar{\mu}^D_{b,t,\omega}(\overline{D}_{b,t} - d_{b,t,\omega}) = 0, \forall b, \forall t \quad (47)$$

complementary slackness

generator & demands

$$\underline{\alpha}^{\text{dis}}_{h,t,\omega} q^{\text{dis}}_{h,t,\omega} = 0, \forall h, \forall t \quad (48)$$

$$\bar{\alpha}^{\text{dis}}_{h,t,\omega}(dis_{h,t} - q^{\text{dis}}_{h,t,\omega})0, \forall h, \forall t \quad (49)$$

$$\underline{\alpha}^{\text{chs}}_{h,t,\omega} q^{\text{chs}}_{h,t,\omega} = 0, \forall h, \forall t \quad (50)$$

$$\bar{\alpha}^{\text{chs}}_{h,t,\omega}(chs_{h,t} - q^{\text{chs}}_{h,t,\omega}) = 0, \forall h, \forall t \quad (51)$$

$$\cancel{\underline{\varphi}_{l,t,\omega}(\overline{F}_l + pf_{l,t,\omega}) = 0, \forall l, \forall t \quad (52)}$$

$$\cancel{\bar{\varphi}_{l,t,\omega}(\overline{F}_l - pf_{l,t,\omega}) = 0, \forall l, \forall t \quad (53)}$$

$$\cancel{\underline{\delta}_{b,t,\omega}(\bar{\theta} + \theta_{b,t,\omega}) = 0, \forall b \ne \text{ref}, \forall t \quad (54)}$$

$$\cancel{\bar{\delta}_{b,t,\omega}(\bar{\theta} - \theta_{b,t,\omega}) = 0, \forall b \ne \text{ref}, \forall t \quad (55)}$$

charging & discharging

$$\beta^u_{i,t,\omega}(RU_i - (p^G_{i,t,\omega} - P^G_{i,0,\omega})) = 0, t = 1, \forall i \quad (56)$$

$$\beta^d_{i,t,\omega}(RD_i - (P^G_{i,0,\omega} - p^G_{i,t,\omega})) = 0, t = 1, \forall i \quad (57)$$

$$\beta^u_{i,t,\omega}(RU_i - (p^G_{i,t,\omega} - p^G_{i,t-1,\omega})) = 0, t = 2 \ldots N_T, \forall i \quad (58)$$

$$\beta^d_{i,t,\omega}(RD_i - (p^G_{i,t-1,\omega} - p^G_{i,t,\omega})) = 0, t = 2 \ldots N_T, \forall i \quad (59)$$

Ramping Gen

$$\underline{\mu}^G_{i,t,\omega}, \bar{\mu}^G_{i,t,\omega}, \underline{\mu}^W_{w,t,\omega}, \bar{\mu}^W_{w,t,\omega}, \underline{\mu}^D_{b,t,\omega}, \bar{\mu}^D_{b,t,\omega},$$

$$\beta^u_{i,t,\omega}, \beta^d_{i,t,\omega}, \underline{\alpha}^{\text{dis}}_{h,t,\omega}, \bar{\alpha}^{\text{dis}}_{h,t,\omega}, \underline{\alpha}^{\text{chs}}_{h,t,\omega}, \bar{\alpha}^{\text{chs}}_{h,t,\omega},$$

$$\cancel{\underline{\varphi}_{l,t,\omega}, \bar{\varphi}_{l,t,\omega}, \underline{\delta}_{b,t,\omega}, \bar{\delta}_{b,t,\omega} \ge 0 \quad (60)}$$

duals P

$$\underline{\mu}^{G}_{i,t,\omega}, \overline{\mu}^{G}_{i,t,\omega}, \underline{\mu}^{W}_{w,t,\omega}, \overline{\mu}^{W}_{w,t,\omega}, \underline{\mu}^{D}_{b,t,\omega}, \overline{\mu}^{D}_{b,t,\omega},$$

$$\beta^{u}_{i,t,\omega}, \beta^{d}_{i,t,\omega}, \underline{\alpha}^{\text{dis}}_{h,t,\omega}, \overline{\alpha}^{\text{dis}}_{h,t,\omega}, \underline{\alpha}^{\text{chs}}_{h,t,\omega}, \overline{\alpha}^{\text{chs}}_{h,t,\omega},$$

$$\geq 0 \quad (60)$$

$$\lambda_{b,t,\omega}: \text{free} \quad (61)$$

duals ?